



2.1 Linear models

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A linear model $y=mx+c$ describes a constant rate of change: m is the gradient (change in y per unit increase in x) and c is the y -intercept (value of y when $x=0$). Linear models are built from a rate and a fixed/starting value, or fitted to two data points, and used to predict values within a sensible domain for the context. Extrapolating far beyond the data used to build the model carries risk, since the true relationship may not stay linear.

Worked example: follow the reasoning

A plumber charges a callout fee of \$45 plus \$30 per hour worked. (a) Write a linear model $C(h)$ for the total cost in terms of hours worked h . (b) Find the cost of a 3.5 hour job.
Solution: (a) $C(h)=45+30h$. (b) $C(3.5)=45+30(3.5)=45+105=\150 .

Quick check · multiple choice

A line passes through (2,7) and (5,16). What is its gradient?

- A 2
- B 3
- C 4.5
- D 9

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A car rental company charges a fixed fee of \$60 plus \$0.25 per km driven. (a) [5]
Write a linear model $C(d)$ for the total cost in terms of distance driven, d km. (b) A customer's bill is \$145. Find the distance driven. (c) State a suitable restriction on the domain of d in this context.

Final answer: _____



Answers and explanation

Question 1 [5]

(a) $C(d)=60+0.25d$. (b) $145=60+0.25d \rightarrow 85=0.25d \rightarrow d=340$ km. (c) $d \geq 0$ (realistically also bounded above by the vehicle's range/rental period).

Quick check answer

B. Gradient = $(16-7)/(5-2) = 9/3 = 3$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.